

Question				Marking details	Marks available																				
					AO1	AO2	AO3	Total	Maths	Prac															
	(b)			award (1) for any of following extraction (of aluminium) is very expensive extraction is much more expensive than recycling recycling is much cheaper than extraction award (1) for reasoning because much larger amounts of electricity/energy needed award (1) for reference to conserving the ore if no other mark awarded	2			2																	
	(c)	(i)		<table><tr><th>Left-hand side of the membrane</th><th>Right-hand side of the membrane</th><th></th></tr><tr><td>Na⁺ Cl⁻ H⁺ OH⁻</td><td>Cl⁻ H⁺ OH⁻</td><td>☐</td></tr><tr><td>H⁺ OH⁻ Cl⁻</td><td>Na⁺ H⁺ OH⁻</td><td>☐</td></tr><tr><td>Na⁺ Cl⁻ H⁺ OH⁻</td><td>Na⁺ H⁺ OH⁻</td><td>☒</td></tr><tr><td>Na⁺ Cl⁻</td><td>Na⁺ OH⁻</td><td>☐</td></tr></table>	Left-hand side of the membrane	Right-hand side of the membrane		Na ⁺ Cl ⁻ H ⁺ OH ⁻	Cl ⁻ H ⁺ OH ⁻	☐	H ⁺ OH ⁻ Cl ⁻	Na ⁺ H ⁺ OH ⁻	☐	Na ⁺ Cl ⁻ H ⁺ OH ⁻	Na ⁺ H ⁺ OH ⁻	☒	Na ⁺ Cl ⁻	Na ⁺ OH ⁻	☐			1	1		
Left-hand side of the membrane	Right-hand side of the membrane																								
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Na ⁺ Cl ⁻ H ⁺ OH ⁻	Na ⁺ H ⁺ OH ⁻	☒																							
Na ⁺ Cl ⁻	Na ⁺ OH ⁻	☐																							
		(ii)		titanium burns in chlorine in aqueous conditions ☐ titanium doesn't react with chlorine under any conditions ☐ aqueous conditions prevent chlorine from reacting with titanium ☒ aqueous conditions prevent chlorine from reacting with sodium chloride ☐			1	1																	